Near-Lossless Long Context under Fixed VRAM via Critical-Span Retention

Portfolio / workshop-style draft
Lab: RTX 3090 24 GB Primary: `Qwen/Qwen3-4B-Instruct-2507`
Repo: `nearlossless-context` Updated: 2026-07-16

Abstract

Long-context inference is limited by KV-cache memory. Training-free eviction methods shrink the cache, but it is often unclear *which* tokens are necessary for near–full-KV quality. On a multi-seed controlled retrieval suite (Qwen3-4B, RTX 3090), we show three linked results:

1. **Structure (H1′).** Retaining critical fact tokens plus a small local radius $R^* = 1$ (with attention sinks and a recent/question window) is **necessary and sufficient** for $\varepsilon \approx 0$ single-fact recall: oracle **15/15**, anti-oracle **0/15**, full KV **15/15** (5 seeds $\times$ 3 depths @4k).
2. **Discovery gap.** Online streaming at a moderate peak budget fails under multi-seed hay not because the budget is too small, but because **query-unknown discovery is weak**: attention stream@512 is **33%**, while perfect online pin of critical$\pm R$ is **15/15** at the same peak.
3. **Closing the gap.** A **surface novelty detector** (within-prefix rarity + digit/ID-like cues; no final question) with **sticky pin packing** reaches $\sim$**93%** multi-seed @4k and **100%** (3×3) through **40k** at stream@512—peak cache $\sim$**1k tokens**, flat in L.

The critical-span mechanism transfers across Qwen2.5, Llama-3.2, and hybrid Gemma-4 (full layers). We do **not** claim general long-context SOTA; see Section 6.

1 Introduction

KV cache memory grows with context length L. On a fixed workstation (here: 24 GB), full-cache decode becomes the bottleneck long before "the model cannot attend." Training-free compressors (observation-window scoring, recent-only eviction, SnapKV-style selection) reduce tokens kept, but practice often optimizes average scores rather than **guaranteeing** the spans that encode answers.

We study **near-lossless** compression on retrieval-critical prompts: quality Q should match full KV at $\varepsilon \approx 0$ under a multi-seed protocol. The systems objective is

$$L_\varepsilon \;=\; \max\bigl\{\, L : Q(M, L) \geq (1-\varepsilon)\, Q(\mathrm{Full}, L) \,\bigr\} \tag{1}$$

under **peak** cache / VRAM constraints—not only final decode size after a full prefill.

Claim. Near-lossless training-free KV compression is better understood as **retaining critical local neighborhoods** (and discovering them online before the question exists) than as uniform thinning or posthoc score ranking alone.

2 Related work (sketch)

- **Eviction / selection:** H2O, SnapKV, pyramid and related training-free keepers.
- **Hybrid attention:** sliding-window + full layers (e.g. Gemma-4); full-layer KV still dominates compressible state.
- **Quantization** (KIVI, etc.): complementary; we use fake-int8 only for logical byte accounting.

Positioning. Mechanism + workstation L_ε, not a leaderboard chase. Novelty is a **query-unknown discovery** layer on top of H1′ retention, not a new attention kernel.

3 Methods (short)

Task. Needle-in-haystack style secrets in seeded filler; depths $\{0, 0.5, 1\}$; success = exact key recall (greedy decode).

H1′ keep set. sinks $\cup$ critical$\pm R$ $\cup$ recent/question window. Kill arms: full, oracle_r1, anti_oracle.

Posthoc scorer. `seed_valley`: shared obs-window attention $\to$ seeds $\to$ contiguous valley + $\pm R$.

Streaming. Chunked prefill; compress when cache $>$ budget; peak $\approx$ budget + chunk.

Oracle-online upper bound. Force-keep critical$\pm R$ whenever still in cache (perfect discovery).

Surface novelty (query-unknown). Per-token score from within-prefix rarity, first occurrence, and surface cues (digits, ID-like pieces). Expand peaks $\pm R$ into pin sets. **Sticky** registry: once discovered, pins stay until evicted; packing ranks pins by score under budget (sinks + recent reserved).

Default API. `prefill_auto(..., mode="stream", discovery="novelty")`.

4 Results

4.1 Figure 1 — Story in four panels

A. H1′ kill (4k, 5×3 multi-seed). Full and oracle crit$\pm$1 both **15/15**; anti-oracle **0/15**. Bare critical tokens without radius often corrupt trailing digits (H1 needs local context); $R^* = 1$ restores $\varepsilon \approx 0$. Mean $|\text{oracle}| \approx$ **149** tokens vs full $\sim$4k.

B. Discovery gap (stream@512). Attention valley **33%**; surface novelty **93%** (14/15); oracle pin **100%**. Peak budget is in the same class ($\sim$1k with chunk). Failures of valley at 512 are **detector** failures, not "stream cannot work."

C. Long L multi-seed (3×3). Sticky novelty@512 is **9/9** at 16k, 24k, 32k, and **40k**. Non-sticky novelty degraded at 24k ($\sim$6/9) via re-rank thrash; sticky packing fixed it. Valley@512 remains end-depth-only ($\sim$33%) under multi-seed hay as L grows.

D. Peak cache. Full KV scales with L; sticky novelty stream@512 stays $\sim$**1k** tokens. Prior valley operating points often used $\sim$1.5–2k peak for long L.

4.2 Scorer tax (posthoc, question known)

Table 1: Posthoc budget tax vs oracle keep set (4k multi-seed).

Method	All-cell $B_{\min}$	Tax vs mean $|\text{oracle}|$
oracle_r1	$\sim$149	1.0$\times$
seed_valley	**192**	$\sim$**1.24**$\times$ global / $\sim$**1.16**$\times$ mean

Near-lossless KV: critical spans → discovery gap → sticky novelty

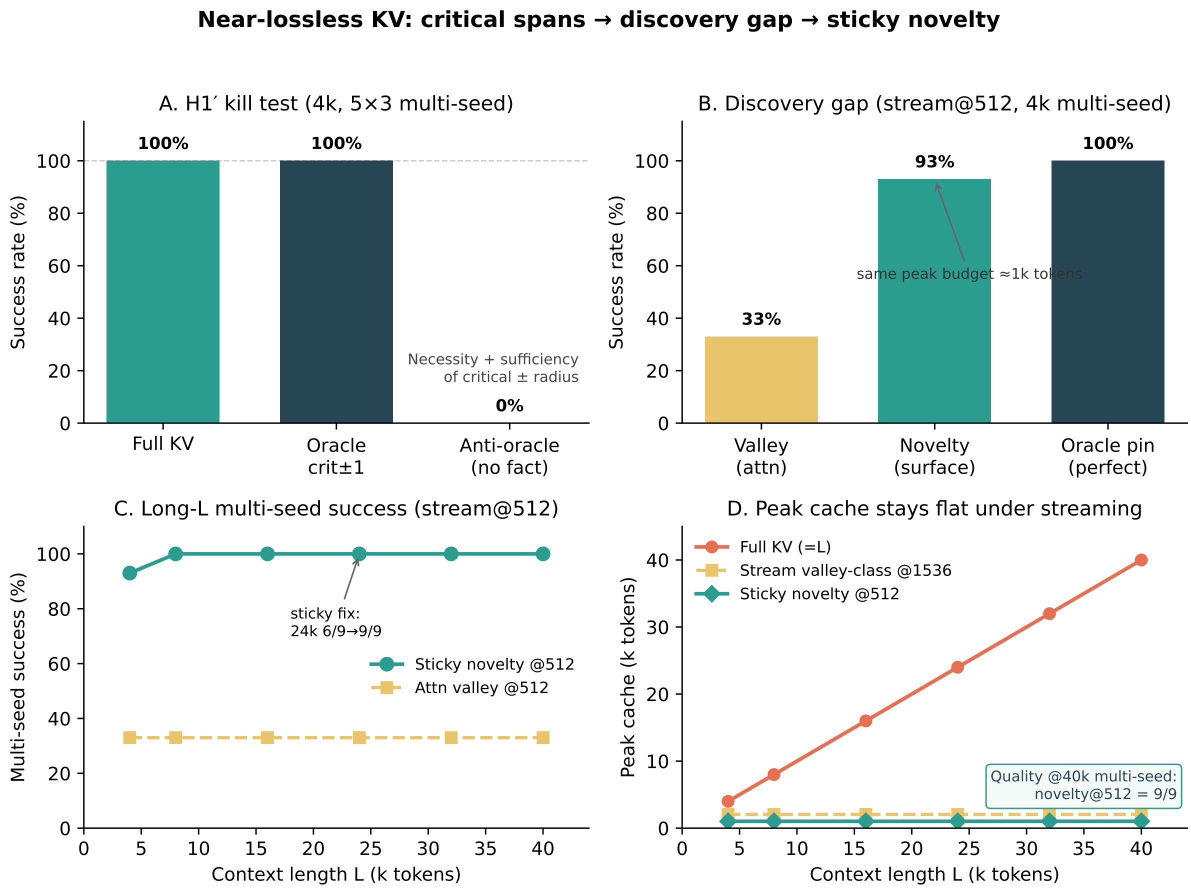

Figure 1: **Story figure.** **(A)** H1′ kill test @4k multi-seed (5×3): full and oracle crit±1 both 15/15; anti-oracle 0/15. **(B)** Discovery gap at stream@512: valley 33%, surface novelty 93%, oracle pin 100%. **(C)** Long-L multi-seed success: sticky novelty@512 holds 9/9 through 40k; valley@512 stays end-only (∼33%). **(D)** Peak cache tokens stay flat (∼1k) under sticky novelty stream@512 while full KV grows with L.

Posthoc is near-oracle-tight; **stream is not**, until discovery improves.

4.3 Stress and transfer (summary)

5 Adaptive policy (systems)

`prefill_auto` applies L-based budgets + per-model floors. With default `discovery="novelty"`, single-needle stream@**512** is the measured multi-seed operating point through **40k**. Attn-only discovery still needs larger floors (∼1536 multi-seed @4k). Hybrid Gemma: compress **full** layers only; sliding layers keep absolute prefix bookkeeping for score-pass.

6 Limitations and what we do *not* claim

6.1 What we claim

- On this **controlled multi-seed retrieval suite**, critical$\pm R^*$ is necessary/sufficient for $\varepsilon \approx 0$ single-fact recall.
- Stream failures at moderate budget are primarily a **query-unknown discovery** problem (oracle-online upper bound).

Table 2: Stress suite and cross-model transfer (summary).

Check	Result
NL facts (names/places) sticky novelty@512	**15/15** multi-seed
Code needles sticky	**15/15**
Adversarial ID-like filler (pre-sticky)	novelty **100%** vs valley **33%**
Multi-needle recall_all @512	novelty $\sim$**80%** (open packing edge)
Gemma-4 E4B hybrid novelty@512	multi-seed **9/9** @4k (valley needs $\sim$1024)
Qwen2.5 / Llama-3.2	H1 holds; family-specific posthoc floors

- Sticky surface novelty **closes most of that gap** through **40k** multi-seed at peak cache $\sim$1k on the primary model, with transfer smoke to other small instruct models including hybrid Gemma-4.

6.2 What we do *not* claim

Table 3: Scope boundaries (honest).

Not claimed	**Why**
General long-context SOTA	No full RULER / LongBench / InfiniteBench leaderboard evaluation.
All task types	Suite is retrieval / needle-class (+ limited multi-needle and NL facts). Reasoning chains, code repos, multi-doc QA largely untested.
Oracle-tight online budgets for free	Sticky novelty $\approx$ oracle quality on this suite at@512, but multi-secret packing ($\sim$80% multi3) and suite-alignment remain.
Production memory stack	Fake-int8 is logical accounting only; no fused kernels or serving productization.
Novelty as universal "importance"	Exploits surface distinctness vs repetitive filler; ID-like filler / filler-like secrets can still confuse discovery.
Large models / training-free SOTA	Primary evidence is $\sim$3–4B instruct models on one GPU class.
Statistical finality	Multi-seed N modest (5×3 @4k; 3×3 long-L). Lab-grade, not a large meta-analysis.
$\varepsilon = 0$ beyond measured envelope	Sticky multi-seed measured through **40k**; longer L is extrapolation.
Peak VRAM = final decode KV	Streaming caps **peak cache tokens**; weights and activations still dominate total VRAM.

6.3 Threats to validity

- **Greedy decode** and chat templates may interact with exact-match scoring.
- **Filler distribution** is synthetic; natural corpora may change novelty baselines.
- **Hybrid models:** only full layers are compressed; sliding-window behavior is infrastructure-sensitive.
- **Selection bias toward positive arms:** we iterated methods until discovery worked; negative results (scored capsules alone, non-sticky long-L) are reported but the path is research-in-the-loop.

7 Conclusion

Near-lossless training-free KV compression on retrieval tasks is structured by **critical local neighborhoods**. When the question is known, a simple attention valley scorer approximates the oracle keep set with small tax. When the question is **not** known—as in online streaming—the binding constraint is **discovery**.

Sticky surface novelty is a lightweight query-unknown detector that, on our multi-seed suite, restores high success through **40k** context at a **flat ∼1k-token peak cache**, transferring to hybrid Gemma-4 at primary-class budgets. The honest next steps are broader tasks and multi-entity packing—not larger single-needle grids alone.

Reproducibility

```
# Multi-seed H1 + scorer tax
python experiments/bench_paper_rigor.py --seeds 0,1,2,3,4 --ctx 4096

# Discovery gap / novelty
python experiments/bench_capsules.py --novelty --oracle-online \
  --skip-capsules --budgets 512
python experiments/bench_novelty_stress.py
python experiments/bench_novelty_longL.py \
  --ctx 16384,24576,32768,40960 --arms novelty --budgets 512

# Figure
python experiments/plot_paper_figures.py

# Transfer
python experiments/bench_transfer.py --model google/gemma-4-E4B-it
```

Primary model: `Qwen/Qwen3-4B-Instruct-2507`, transformers, CUDA bf16, RTX 3090.
Artifact index: `results/FINDINGS.md`.
Markdown twin: `papers/PAPER_DRAFT.md`.
Build PDF: from `papers/`, run `powershell -File build_pdf.ps1`.